

Econ 101A

Section 7

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1. Expenditure Minimization

Sometimes we maximize utility subject to a budget constraint. Other times we minimize expenditure subject to a utility constraint. Here are the relevant details.

	Utility Maximization	Expenditure Minimization
Problem	$\max_{x_1, x_2} U(x_1, x_2) \text{ s.t. } p_1 x_1 + p_2 x_2 - M \leq 0$	$\min_{x_1, x_2} p_1 x_1 + p_2 x_2 \text{ s.t. } U(x_1, x_2) \geq U_0$
Lagrangian	$\mathcal{L}(x_1, x_2, \lambda) = U(x_1, x_2) - \lambda (p_1 x_1 + p_2 x_2 - M)$	$\mathcal{L}(x_1, x_2, \lambda) = p_1 x_1 + p_2 x_2 - \lambda U(x_1, x_2 - U_0)$
FOC	$\frac{U'_{x_1}}{U'_{x_2}} = \frac{p_1}{p_2}, \quad \lambda = \frac{U'_{x_i}}{p_i}$	$\frac{U'_{x_1}}{U'_{x_2}} = \frac{p_1}{p_2}, \quad \lambda = \frac{p_i}{U'_{x_i}}$
SOC	$ H_B > 0$	$ H_B < 0$
Demand Name	Marshallian demand	Hicksian (compensated) demand
Demand Equations	$x_j^*(p_1, p_2, M)$	$h_j^*(p_1, p_2, U_0)$
Indirect utility function	$v(p_1, p_2, M) = U(x_1^*(p_1, p_2, M), x_2^*(p_1, p_2, M))$	
Expenditure function		$e(p_1, p_2, U_0) = p_1 h_1^*(p_1, p_2, U_0) + p_2 h_2^*(p_1, p_2, U_0)$

- If we set U_0 in the expenditure-minimization problem equal to optimized value of the objective function in the corresponding utility-maximization problem (i.e. equal to the indirect utility function), we will get $(h_1^*, h_2^*) = (x_1^*, x_2^*)$. Similarly, if we set M in the utility-maximization problem equal to the optimized value of the objective function in the corresponding expenditure-minimization problem (i.e. equal to the expenditure function e), we will also get $(x_1^*, x_2^*) = (h_1^*, h_2^*)$.
- In other words: $x^*(\mathbf{p}, e(\mathbf{p}, U_0)) = h^*(\mathbf{p}, U_0)$ & $h^*(\mathbf{p}, v(\mathbf{p}, M)) = x^*(\mathbf{p}, M)$. We sometimes call this the **dual property**.
- **Warning:** The values are equal, but the functional forms are not. Remember, x^* is the Marshallian demand function and is a function of $\mathbf{p}$ and M while h^* is the Hicksian demand function and is a function of $\mathbf{p}$ and U_0 .
- For both of these problems, we ignore the non-negativity constraints and solve for the interior solution. If the solution proves to be negative for some variables, then we check the corners (all of them!) for the optimum.

2. Useful properties of Hicksian demand function

1. Dual properties: $x^*(\mathbf{p}, e(\mathbf{p}, U_0)) = h^*(\mathbf{p}, U_0)$ & $h^*(\mathbf{p}, v(\mathbf{p}, M)) = x^*(\mathbf{p}, M)$
Similarly: $e(\mathbf{p}, v(\mathbf{p}, M)) = M$ (The expenditure function in emin problem, when utility level is set at the indirect utility of umax problem, exactly equals the income in umax problem)

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2. The expenditure function $e(\mathbf{p}, U_0)$ is concave in $\mathbf{p}$.

Proof. This is equivalent to proving the following: For any $\mathbf{p}$, $\mathbf{p}'$, and $t \in (0, 1)$,

$$e(t\mathbf{p} + (1-t)\mathbf{p}', U_0) \geq te(\mathbf{p}, U_0) + (1-t)e(\mathbf{p}', U_0)$$

Suppose the solution to the emim problem given $\mathbf{p}$ is $\mathbf{h}$, the solution given $\mathbf{p}'$ is $\mathbf{h}'$, and the solution given the convex price $t\mathbf{p} + (1-t)\mathbf{p}'$ is $\mathbf{h}''$. One step back to the definition of $e(\mathbf{p}, U_0)$: This is the minimal expenditure that guarantees utility level U_0 ; $\mathbf{h}$ is the consumption level that leads to the minimal expenditure. Therefore, if you consume a different bundle, say $\mathbf{h}''$, then under the same price $\mathbf{p}$, the expenditure will be higher or at least the same. Mathematically,

$$\mathbf{p} \cdot \mathbf{h}'' \geq \mathbf{p} \cdot \mathbf{h} = e(\mathbf{p}, U_0) \quad (1)$$

Please try to apply the same logic to prove the following step:

$$\mathbf{p}' \cdot \mathbf{h}'' \geq \mathbf{p}' \cdot \mathbf{h}' = e(\mathbf{p}', U_0) \quad (2)$$

Combine these two equations, multiplied by t or $(1-t)$, you get what we want to establish:

$$\begin{aligned} e(t\mathbf{p} + (1-t)\mathbf{p}', U_0) &= (t\mathbf{p} + (1-t)\mathbf{p}') \cdot \mathbf{h}'' \\ &= t\mathbf{p} \cdot \mathbf{h}'' + (1-t)\mathbf{p}' \cdot \mathbf{h}'' \\ &\geq t\mathbf{p} \cdot \mathbf{h} + (1-t)\mathbf{p}' \cdot \mathbf{h}' \\ &= te(\mathbf{p}, U_0) + (1-t)e(\mathbf{p}', U_0) \end{aligned}$$

3. Shephard's lemma: $\frac{\partial e(\mathbf{p}, U_0)}{\partial p_i} = h_i^*(\mathbf{p}, U_0)$. [Idea: Apply Envelope Theorem; see lecture notes]

4. Hicksian demand function h_i^* is decreasing in p_i : $\frac{\partial h_i^*(\mathbf{p}, U_0)}{\partial p_i} < 0$ (Recall that this is almost always true for Marshallian demand as well, but there are exceptions, namely Giffen good.)

Proof. Since expenditure function is concave in $\mathbf{p}$, $\frac{\partial^2 e(\mathbf{p}, U_0)}{\partial p_i^2} < 0$. Use Shephard's lemma,

$$\frac{\partial^2 e(\mathbf{p}, U_0)}{\partial p_i^2} = \frac{\partial}{\partial p_i} \left(\frac{\partial e(\mathbf{p}, U_0)}{\partial p_i} \right) = \frac{\partial}{\partial p_i} h_i^*(\mathbf{p}, U_0) < 0$$

5. Mathematical proof of Slutsky equation (we'll focus on intuition in next section): Starting from the dual property, without loss of generality, assuming a society with two goods,

$$h_i^*(p_1, p_2, U_0) = x_i^*(p_1, p_2, e(p_1, p_2, U_0))$$

Take the partial of p_i on both sides, use chain rule on the right hand side,

$$\begin{aligned} \frac{\partial h_i^*(p_1, p_2, U_0)}{\partial p_i} &= \frac{\partial x_i^*(p_1, p_2, e)}{\partial p_i} + \frac{\partial x_i^*(p_1, p_2, e)}{\partial M} \underbrace{\frac{\partial e(p_1, p_2, U_0)}{\partial p_i}}_{\text{Shephard's rule}} \\ &= \frac{\partial x_i^*(p_1, p_2, e)}{\partial p_i} + \frac{\partial x_i^*(p_1, p_2, e)}{\partial M} h_i^*(p_1, p_2, U_0) \end{aligned}$$

Rewrite the equation in terms of Marshallian demand (because it's easier to compute); basically you set $M = e(p_1, p_2, U_0)$, and

$$\frac{\partial x_i^*(p_1, p_2, M)}{\partial p_i} = \underbrace{\frac{\partial h_i^*(p_1, p_2, U_0)}{\partial p_i}}_{\substack{\text{Substitution Effect} \\ < 0}} - \underbrace{x_i^*(p_1, p_2, M) \frac{\partial x_i^*(p_1, p_2, M)}{\partial M}}_{\substack{\text{Income Effect} \\ < 0 \text{ if normal} \\ > 0 \text{ if inferior}}$$

3. Exercises

3.1 Expenditure Minimization

Consider a consumer with Cobb-Douglas preferences:

$$u(x_1, x_2) = \alpha \log(x_1) + (1 - \alpha) \log(x_2)$$

1. Derive the Hicksian demand functions: $h_i^*(p_1, p_2, U_0)$.
2. Derive the expenditure function: $e(p_1, p_2, U_0)$.
3. Derive the Marshallian demand functions: $x_i^*(p_1, p_2, M)$.
4. (Optional) Derive the indirect utility function: $v_i(p_1, p_2, M)$.
5. Verify that $x_i^*(p_1, p_2, e(p_1, p_2, U_0)) = h_i^*(p_1, p_2, U_0)$.

3.2 Utility Maximization & Indirect Utility Functions (Midterm Fall 2005)

Consider Mario, a Californian resident with income M . Mario cares about own consumption c and about a charitable project G in Louisiana. The welfare effects of the project for Louisiana citizens are measured by $h(g + S)$, where g is Mario's charitable donation and S is the seed money donated by others. Mario derives utility $\alpha h(g + S)$ from the project, with $\alpha \geq 0$. Therefore Mario maximizes

$$\begin{aligned} \max_{c, g} & u(c) + \alpha h(g + S) \\ \text{s.t.} & c + g \leq M, c \geq 0, g \geq 0 \end{aligned}$$

We assume $u'(\cdot) > 0$, $u''(\cdot) < 0$, $h'(\cdot) > 0$, and $h''(\cdot) < 0$. That is, both u and h are increasing, concave functions. The function h is concave because there are diminishing returns to the charitable project. We also assume $\alpha, S, M \geq 0$.

1. What is the interpretation of α in the utility function? Interpret in particular the special case $\alpha = 0$.
2. Argue that the problem can be rewritten as

$$\begin{aligned} \max_g & u(M - g) + \alpha h(g + S) \\ \text{s.t.} & 0 \leq g \leq M \end{aligned}$$

3. Neglecting the constraint $0 \leq g \leq M$ for now, derive the first- and second-order conditions. Argue that the point identified by the first-order conditions is a maximum.
4. Solve for g^* and c^* using the first-order conditions for the case $u(c) = \log(c)$ and $h(g) = \log(g)$.
5. Keep assuming $u(c) = \log(c)$ and $h(g) = \log(g)$. Now it is time to check the constraint $0 \leq g \leq M$. Is the constraint $g \leq M$ always satisfied? Is the constraint $g \geq 0$ always satisfied? If not, provide an example to the converse. Write the solution for g^* and c^* , taking into account corner solutions.
6. Keep assuming $u(c) = \log(c)$ and $h(g) = \log(g)$. How do c^* and g^* depend on M , α , and S ? Provide intuition for each of these comparative statics, in particular for the latter.
7. Now we go back to the general formulation with $u(\cdot)$ and $h(\cdot)$ concave. Using the implicit function theorem, and neglecting corner solutions, show that $\partial g^* / \partial S < 0$. Comment on this result.

8. (Open-ended) An economist, John List, does an empirical study of the effect of seed money (the money collected early in a fundraising drive) on charitable donations. He finds that higher seed money is associated with *higher* donations, against the prediction in point 7. How can we explain this finding?
9. (Optional) Keep assuming $u(\cdot)$ and $h(\cdot)$ concave. Consider the indirect utility $V(S, M, \alpha) = u(M - g^*(S, M, \alpha)) + \alpha h(g^*(S, M, \alpha) + S)$. Use the envelope theorem to compute $dV(S, M, \alpha)/dS$, that is, how the indirect utility vary as the seed money increases. Under what conditions is $\partial V(S, M, \alpha)/\partial S > 0$ and why?